

Environmental Noise Analysis Report

Device / Location: Monitoring location (identifier withheld)
Source File: Monitoring location (identifier withheld)
Report Type: COMPREHENSIVE | Generated: 2026-10-10 16:15:46

Section 1: Executive Acoustic Summary

Device / Location ID: Monitoring location (identifier withheld)

Source File: Monitoring location (identifier withheld)

Measurement Date Range: 2026-04-06 00:00:00 to 2026-04-09 23:59:59

Time basis: US Eastern, with daylight saving (EST/EDT)

Total Duration: 4.00 days (96.00 hours)

4-Day Energy Average (LAeq): 51.29 dB(A)

The equivalent continuous sound level (LAeq) over the measurement period was 51.29 dB(A). Section 3 compares the record with the WHO 2018 guideline values and the Maryland COMAR limits.

Non-Technical Summary: Noise Exposure & Health Assessment

This dataset covers 4 days of outdoor noise monitoring from 06 Apr 2026 to 09 Apr 2026. 1 interruption with no readings was detected, totalling 2 h; data completeness was 97.91%. All times are US Eastern, with daylight saving (EST/EDT).

The 4-day energy-average level (LAeq) was 51.3 dB(A). Daytime levels (07:00–22:00) averaged 52.7 dB(A) and nighttime levels (22:00–07:00) averaged 46.4 dB(A). Daytime exceeded nighttime by 6.2 dB. The highest single sound level recorded (L-Max) was 81.0 dB(A).

The level exceeded 10% of the time (L10) was 54.6 dB(A) and the level exceeded 90% of the time (L90) was 43.6 dB(A), a spread of 11.1 dB.

Comparison with the WHO 2018 road-traffic guideline values (total measured level, all sources):

- 24-hour weighted average (Lden): 54.6 dB(A). Above the WHO 2018 road-traffic guideline value of 53.0 dB(A) by 1.6 dB.
- Nighttime level (Lnight, 23:00–07:00): 46.2 dB(A). Above the WHO 2018 road-traffic guideline value of 45.0 dB(A) by 1.2 dB.

Top Peak Noise Events

The loudest discrete events in the record. Each event is a contiguous run of LEQ readings at or above the 90th-percentile level (L10); **Peak LEQ** is the highest 1 s LEQ reading in the event and **Peak Time** the moment it was recorded (US Eastern, with daylight saving (EST/EDT)). **Duration** is the number of readings in the event times the logging interval. Listed events are at least 5 minutes apart.

#	Date	Day	Peak Time	Peak LEQ (1 s)	Duration
1	07 Apr 2026	Tuesday	12:27:22	77.8 dB(A)	55 s
2	07 Apr 2026	Tuesday	16:59:29	72.9 dB(A)	1 min 2 s
3	06 Apr 2026	Monday	14:35:21	72.8 dB(A)	55 s
4	08 Apr 2026	Wednesday	14:17:11	72.3 dB(A)	1 min 5 s
5	07 Apr 2026	Tuesday	18:16:16	72.0 dB(A)	33 s
6	07 Apr 2026	Tuesday	16:12:08	71.4 dB(A)	37 s
7	08 Apr 2026	Wednesday	14:59:21	71.1 dB(A)	1 min 6 s
8	07 Apr 2026	Tuesday	13:34:12	70.7 dB(A)	48 s
9	08 Apr 2026	Wednesday	16:12:13	70.3 dB(A)	20 s
10	06 Apr 2026	Monday	17:40:17	70.2 dB(A)	41 s

Section 2: Data Quality & Completeness (QA/QC)

Measurement Span: 2026-04-06 00:00:00 to 2026-04-09 23:59:59

Time basis: US Eastern, with daylight saving (EST/EDT)

Detected Logging Interval: 1 s

Expected Samples: 345,600

Actual Samples in Dataset: 338,400

Data Completeness: 97.91%

Data Continuity Log

1 interruption with no readings (0 shorter than 15 minutes, 1 of 15 minutes or longer), totalling 2 h.

- No readings between 2026-04-08 01:59:59 and 2026-04-08 04:00:00 (2 h).

Section 3: Comparison with Health Guidelines and Regulatory Limits

Standards sourced from WHO Environmental Noise Guidelines (2018), WHO Guidelines for Community Noise (1999), and Maryland COMAR 26.02.03. OSHA/NIOSH occupational standards are excluded — this is an environmental assessment.

Guideline or limit	Metric	Measured (dB(A))	Value (dB(A))	Measured vs value
WHO 2018 — Road Traffic	Lden	54.60	53.00	Above the value by 1.6 dB
WHO 2018 — Road Traffic	Lnight	46.20	45.00	Above the value by 1.2 dB
WHO 2018 — Aircraft Noise	Lden	54.60	45.00	9.6 dB above source reference — indicative only
WHO 2018 — Aircraft Noise	Lnight	46.20	40.00	6.2 dB above source reference — indicative only
WHO 2018 — Railway	Lden	54.60	54.00	0.6 dB above source reference — indicative only
WHO 2018 — Railway	Lnight	46.20	44.00	2.2 dB above source reference — indicative only
MD COMAR Day (07:00–22:00)	LAeq (07:00–22:00)	52.70	65.00	At or below the value by 12.3 dB
MD COMAR Night (22:00–07:00)	LAeq (22:00–07:00)	46.40	55.00	At or below the value by 8.6 dB

A level at or below the Maryland COMAR values does not imply absence of health risk: the WHO 2018 health-based guideline values are lower.

Time and periods above the limits

Maryland COMAR. The measured level was at or above the 65 dB(A) day limit for **0.2%** of measured daytime (07:00–22:00), and at or above the 55 dB(A) night limit for **0.5%** of measured night-time (22:00–07:00). **WHO 2018.** The measured level was at or above the 45 dB(A) Lnight guideline for **54.0%** of the measured WHO night window (23:00–07:00). Lnight exceeded 45 dB(A) on **4 of 4** nights, each computed over its own 23:00–07:00 window. Lden exceeded 53 dB(A) on **4 of 4** days.

On the shares of time. Each share is counted within its own window and against its own level. The COMAR night figure covers 22:00–07:00 against 55 dB(A); the WHO night figure covers 23:00–07:00 against 45 dB(A). Both are plain LAeq comparisons on the measured scale, so both are computable in the same way — they differ in window and in level, not in kind. A night share uses measured night-time as its denominator, never the 24-hour day, which would dilute it with hours the night level does not govern. All are shares of *measured* time, counted on the 1-second readings the logger stored; a shorter logging interval resolves brief peaks that a longer one averages away. **Lden has no such figure** because it adds +5 dB to evening and +10 dB to night readings before averaging: its 53 dB(A) sits on a penalty-weighted scale that no measured reading is on, so a count of readings above 53 dB(A) would not be about Lden at all. **None of these shares is a compliance determination.** Every value here is compared on a period average, and a period can sit below on its average while spending real time above the level; the comparison rows above carry that comparison. WHO further intends Lden and Lnight as long-term annual averages, so figures from a short record are indicative.

How to read these rows: Each row compares the total measured level with a guideline or limit value and states whether it is above, or at or below, that value. It is not a determination of compliance. The WHO 2018 guideline values are specific to one noise source (road traffic Lden 53 dB(A) and Lnight 45 dB(A); aircraft Lden 45 dB(A); railway Lden 54 dB(A)), while the sound level meter measures the combined sound of all sources and cannot attribute it to one; the aircraft and railway rows are therefore shown as **indicative** only. The Maryland COMAR values apply to noise a person causes at a receiving property; motor vehicles on public roads, licensed airports, railroads and residential air-conditioning are exempt (COMAR 26.02.03.02C), and a compliance measurement is made at the receiving property line with a Type II or better meter (26.02.03.02D). Additionally, WHO 2018 intends Lden/Lnight to represent long-term annual average exposure; a measurement period of days or weeks is indicative only.

Section 4: Single-Event Sleep Disturbance (L-Max only)

Nighttime hours isolated: 23:00–07:00. Peak extraction uses **L-Max dB -A** only.

This section uses the WHO night window of 23:00–07:00, which is the basis of the Lnight guideline. The Maryland COMAR night limit assessed in Section 3 is defined over 22:00–07:00, so the two windows differ by one hour by design.

Absolute highest nighttime peak (LAMax): 68.24 dB(A)

Cumulative nighttime exposure exceeding 60.00 dB(A) (outdoor facade): 6.0 minutes (measured logging interval 1 s). 361 of 108,000 nighttime samples (0.3%).

The 60 dB(A) outdoor figure is used because WHO sets its single-event sleep-disturbance guideline indoors, at 45 dB(A) LAMax (WHO Guidelines for Community Noise, 1999; carried forward in the Night Noise Guidelines for Europe, 2009). Converting it to an outdoor facade level assumes roughly 15 dB of attenuation through a partially open window, the value WHO uses for that purpose. Actual attenuation depends on the construction, glazing and window position of the specific dwelling and was not measured here, so this comparison is indicative. A direct indoor measurement is required to establish indoor exposure.

Section 5: Daily Summary Matrix (Multi-Week Variance)

Times are US Eastern, with daylight saving (EST/EDT). LAeq is the energy average of the LEQ readings in each period. The night column is the night that begins on the evening of the date shown; the Lden column covers the 24 hours from 07:00 on that date. * less than 99.5% of the period was measured (coverage in brackets).

Date dB(A)	LAeq 00:00–24:00 dB(A)	LAeq 07:00–22:00 dB(A)	LAeq night 22:00–07:00 dB(A)	Lden 07:00–07:00 dB(A)
2026-04-06 (Mon)	50.75	52.11	45.84	53.99
2026-04-07 (Tue)	51.91	53.54	46.04* (77%)	54.81* (91%)
2026-04-08 (Wed)	51.80* (91%)	53.00	46.76	54.98
2026-04-09 (Thu)	50.55	51.89	48.30* (22%)	55.59* (70%)

Section 6: Diurnal Hourly Profile for 4 days

Hour	Average LAeq (dB(A))	Min (dB(A))	Max (dB(A))	Std Dev (dB)	Hour	Average LAeq (dB(A))	Min (dB(A))	Max (dB(A))	Std Dev (dB)
00:00	47.17	40.06	61.33	1.88	12:00	54.02	44.49	77.85	2.28
01:00	46.83	40.09	61.13	1.98	13:00	54.94	45.69	70.66	2.13
02:00	46.20	40.02	59.64	1.74	14:00	56.06	48.13	72.81	2.36
03:00	45.60	39.03	59.30	1.82	15:00	55.50	48.57	69.99	2.09
04:00	44.53	37.89	57.78	1.87	16:00	54.99	47.25	72.90	2.19
05:00	44.04	37.21	59.90	1.84	17:00	53.80	45.99	70.22	2.19
06:00	45.91	37.69	62.73	2.83	18:00	52.99	44.14	72.05	2.50
07:00	45.71	37.53	62.70	2.52	19:00	51.02	44.15	67.46	2.15
08:00	47.07	38.46	63.47	2.66	20:00	50.35	42.48	66.87	2.38
09:00	47.93	39.51	64.72	2.29	21:00	48.85	41.82	64.10	2.02
10:00	49.80	41.47	67.15	2.20	22:00	48.02	40.57	63.04	1.86
11:00	51.77	44.18	67.95	2.16	23:00	47.77	40.44	63.60	1.95

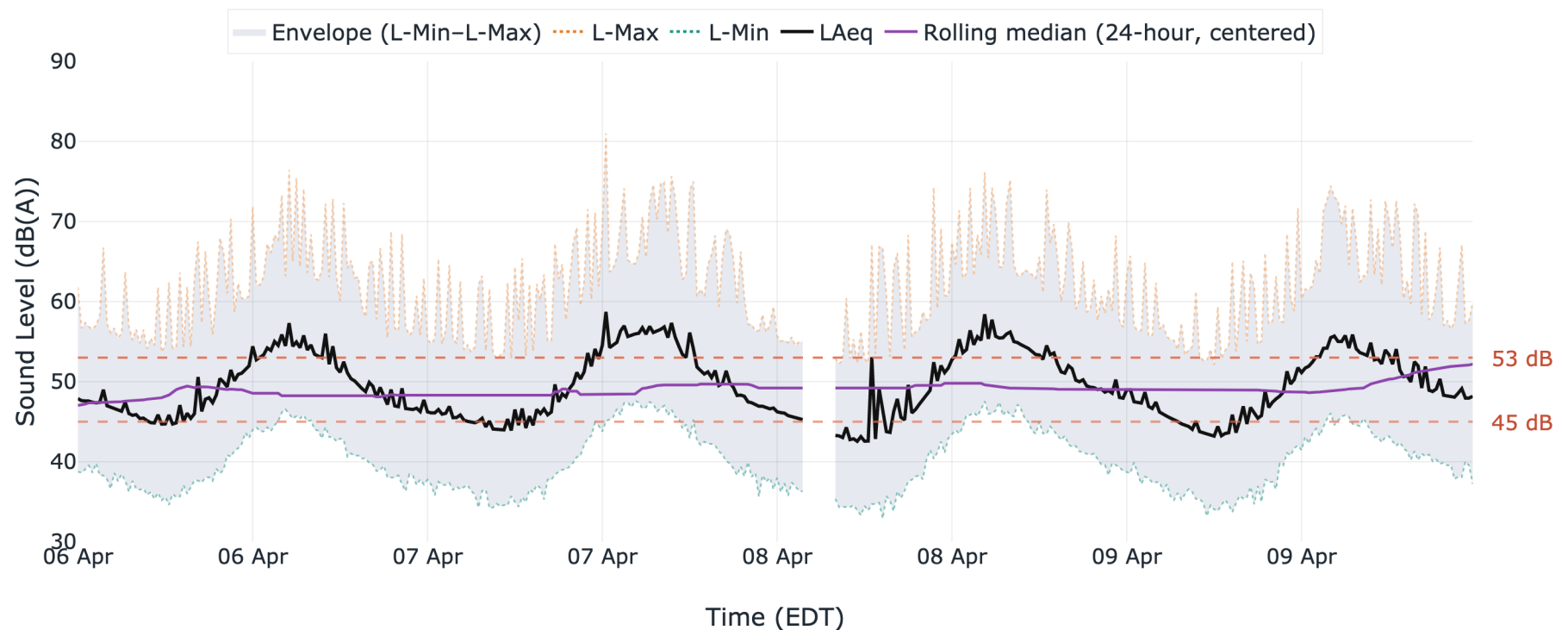
Section 7: Statistical Noise Profile (Percentiles)

Percentile	Value (dB(A))	Definition
L5	55.80	High-noise / transient events (exceeded 5% of time)
L10	54.62	High-noise / transient events (exceeded 10% of time)
L50	48.30	Median acoustic climate (exceeded 50% of time)
L90	43.56	Background / ambient baseline (exceeded 90% of time)
L95	42.58	Background / ambient baseline (exceeded 95% of time)

Section 8: Advanced Visualizations

Chart 1: Time Series (LAeq with L-Max/L-Min envelope & WHO limits)

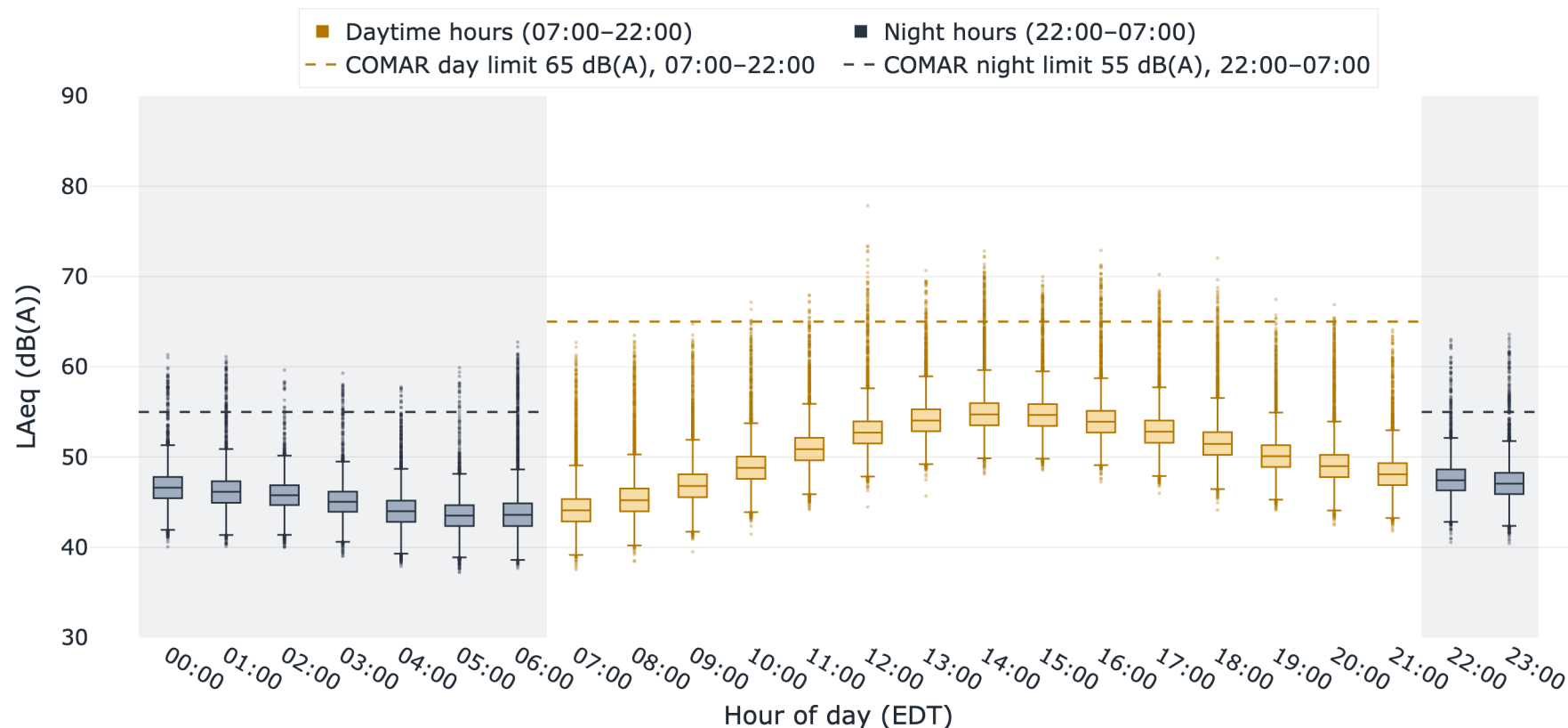
Chart 1: Time Series — LAeq with L-Max/L-Min envelope (15-minute bins)



What it is: LAeq across the full measurement period, with the L-Min to L-Max envelope. **How it is calculated:** **Black line** — LAeq in consecutive 15-minute bins: every sample in a bin is combined by logarithmic energy averaging into one value, giving one point per bin, not one per sample. Bin length is set by record length (15-minute for records up to 15 days, 1-hour for records up to 62 days, and 24-hour beyond that). **Shaded band** — L-Min to L-Max within the same bins. **Purple line** — centered 24-hour moving median, which removes the daily cycle and leaves the underlying trend. **Breaks** — bins with no measurement are left blank, so a gap is an outage, not a quiet period. **Dashed lines** — the WHO 2018 values of 53 and 45 dB(A), drawn as a scale only. WHO sets its limits on Lden and Lnight, whole-period figures reported in Section 3, not on a 15-minute average, so a bin above a line is not by itself an exceedance. **How to read it:** The dark line is the sustained acoustic load; the width of the shaded band is the spread between the quietest and loudest moments within each bin. **Data source:** Monitoring location (identifier withheld).

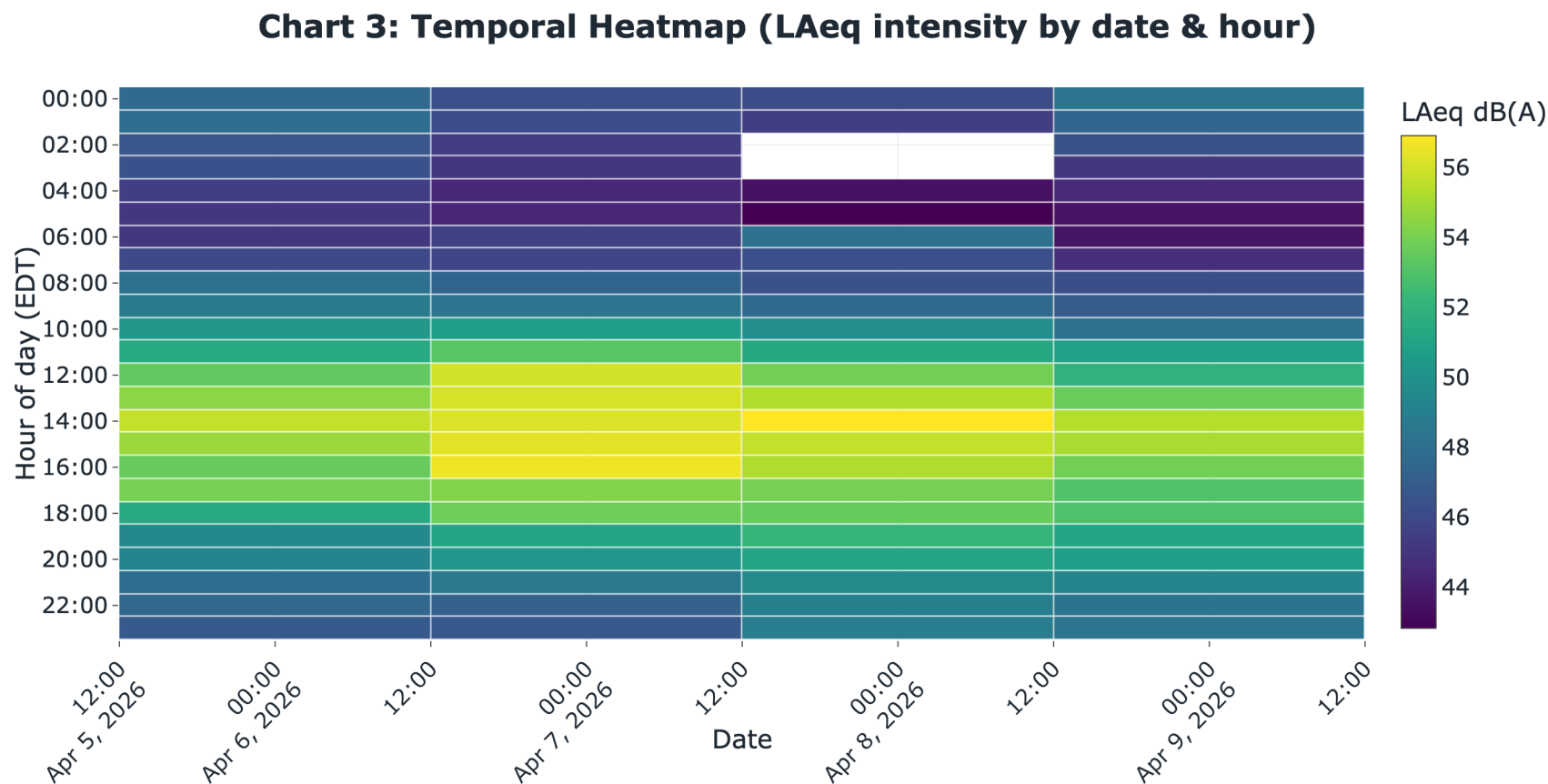
Chart 2: Diurnal Box-and-Whisker (Hourly LEQ Volatility)

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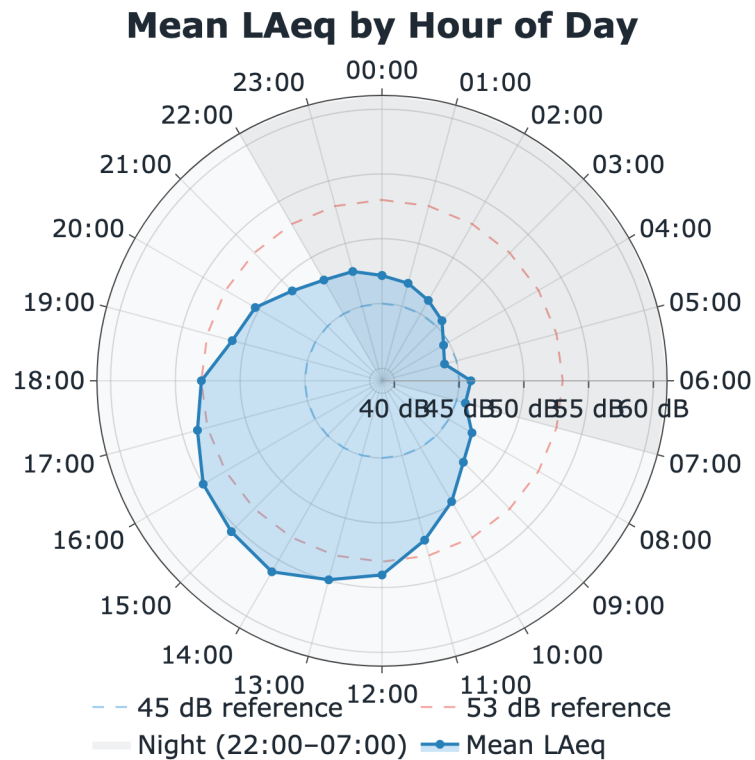
What it is: The full statistical spread of LAeq at each hour of the day, not a single average. **How it is calculated:** A visual summary of the LAeq at each hour of the day, pooled across every measured day: the 00:00 box holds every reading taken between 00:00 and 00:59 on any day of the record. **Box** — the interquartile range (IQR), the middle 50% of readings, from the 25th percentile (Q1) to the 75th (Q3). **Median** — the line across the box: half the readings are louder, half quieter. **Whiskers** — extend to the furthest reading within 1.5 x IQR of the box, one above Q3 and one below Q1. **Outliers** — the points beyond the whiskers: 1-second readings that fall outside the majority of the data, the quietest at night and the loudest by day. What produced any individual outlier cannot be determined from sound level data. A tall box means the level at that hour varied widely between days; a short box means it was consistent. **Colours** — orange is daytime (07:00–22:00), dark blue on the shaded background is night (22:00–07:00), following the Maryland COMAR definition. [3] **Dashed lines** — the COMAR 26.02.03.02B(1) Table 1 residential limits, 65 dB(A) by day and 55 dB(A) by night, each drawn only across the hours its period covers. They apply to the LAeq of the whole day or whole night period, reported in Section 3, not to a single hour or a single reading. [2] **How to read it:** A tall box indicates acoustically variable conditions at that hour; hours with short, low boxes are steady and quiet. What produces either pattern cannot be determined from sound level data alone. **Data source:** Monitoring location (identifier withheld).

Chart 3: Temporal Heatmap (LAeq Intensity by Date & Hour)



What it is: LAeq for every hour of every day in the record. **How it is calculated:** **Cells** — one clock hour on one date, computed by logarithmic energy averaging (never an arithmetic mean) over the samples in that hour; blank cells are hours with no data. **Colour** — viridis: dark purple quietest, green mid-range, bright yellow loudest. Colour reflects the measured level only; a single hour cannot be compared against the WHO guidelines, which are defined on Lden and Lnight. Every second hour is labelled, each tick on the centre of its cell. **How to read it:** Read down a column to see how one day changed hour by hour; read across a row to see whether a given hour behaved the same way from day to day. A consistently bright row indicates a recurring daily pattern; identifying what produces it requires observations beyond sound level data. **Data source:** Monitoring location (identifier withheld).

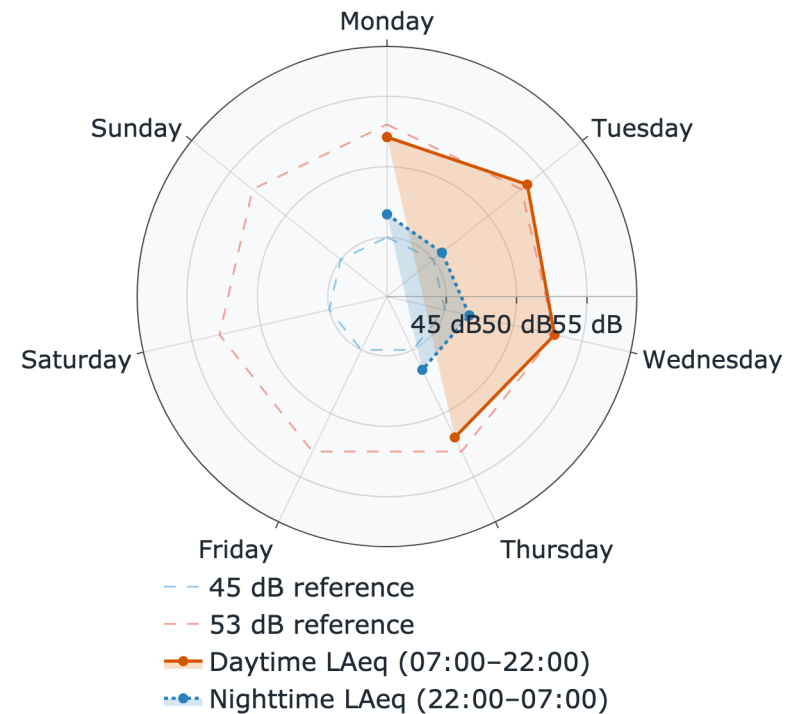
Chart 4: Diurnal Noise Fingerprint (24-Hour Polar Radar)



What it is: A polar radar chart showing the mean LAeq noise level for each of the 24 hours of the day, plotted clockwise from midnight (00:00) around the circle. **How it is calculated:** All measurements falling within each clock hour are energy-averaged (LAeq) across every day in the dataset. Each spoke is labelled by the hour it starts: the 06:00 spoke is the 06:00–06:59 bin. **How to read it:** The shaded sector spans the night period, 22:00 to 07:00 (Maryland COMAR), and so covers the hourly bins from 22:00 through 06:00. The polygon shape summarises the site's daily level pattern. A lopsided polygon with morning and late-afternoon peaks indicates activity concentrated at those hours; a uniformly expanded polygon indicates a level that is broadly steady around the clock. What produces either pattern cannot be determined from sound level data alone and requires corroborating observation. The dashed rings mark 45 dB and 53 dB for visual orientation only. They are the WHO guideline VALUES, but those guidelines are defined on L_{night} and L_{den} — a night-long and a 24-hour penalty-weighted average respectively — so an individual hour rising above a ring is not an exceedance. The comparison in Section 3 uses the correct metrics. **Data source:** Monitoring location (identifier withheld).

Chart 5: Weekly Noise Profile (Day-of-Week Radar)

Weekly Profile — Daytime vs Nighttime LAeq



What it is: A 7-spoke radar chart showing the mean LAeq noise level for each day of the week (Monday–Sunday), with separate traces for daytime (07:00–22:00) and nighttime (22:00–07:00). **How it is calculated:** All measurements are grouped by day-of-week and time-of-day period, then energy-averaged (LAeq) across all occurrences of that combination in the dataset. **How to read it:** A wider daytime polygon shows that daytime levels exceed nighttime levels. Shorter weekend than weekday spokes indicate a level that falls at weekends, and roughly equal spokes indicate a level that does not vary by day of week. These are descriptions of the measured pattern; attributing any of them to a particular source requires evidence beyond sound level data. **Data source:** Monitoring location (identifier withheld).

Section 9: Methodological Limitations & Disclaimer

Data Provenance & Chain of Custody

Source file: Withheld (see SHA-256 below for verification)

SHA-256 of source file: 99430614b36bdb52f5317c3cbaaf31fa8f2303eda0bbd3da82fdd3c8ec99a527

Samples analysed: 338,400

Device / location identifier: Monitoring location (identifier withheld)

Sensor placement: outdoor

Report generated: 2026-10-10 16:15:47

De-identification. This report contains no participant name, address or other personal identifier. Files are referred to by position rather than by filename. The following were withheld: device/location identifier. Verification does not depend on those names: the SHA-256 digest above identifies the analysed file uniquely, so a reader holding the original can confirm it is the file this report was produced from.

Instrument and calibration. Not provided. The data file carries no instrument metadata and none was entered, so the instrument model, serial number and calibration are not stated in this report. The deployment geometry (microphone height, orientation and distance from any reflecting facade) is not recorded in the data file and is not reproduced here. Levels reported here are reproducible from the source file independently of that documentation.

Time Basis and Periods

All times in this report are US Eastern, with daylight saving (EST/EDT). Each period takes readings from its own hours only: a calendar day runs 00:00 to 24:00; the COMAR day 07:00 to 22:00; a night (22:00 to 07:00 for COMAR, 23:00 to 07:00 for Lnight) is labelled with the evening on which it begins; and a daily Lden covers the 24 hours from 07:00, its night being the one that follows (EU Directive 2002/49/EC, Annex I). When daylight saving time begins, the clock moves from 01:59 to 03:00 and that day has 23 hours; when it ends, 01:00 to 01:59 occurs twice and both passes are kept, in order. Neither change is treated as missing or duplicate data, and period coverage is measured against each period's real length.

Averaging Period

The WHO 2018 Lden and Lnight guideline values are defined on a LONG-TERM average, conventionally a full year. The Lden and Lnight reported here are computed over the duration of this measurement only, which is very much shorter. They are therefore an indication of conditions during the monitored period, not a determination of long-term exposure, and a single unusually quiet or unusually busy week will move them. Seasonal variation, weekday/weekend composition and weather during the measurement all affect the result. Comparisons against the WHO values in this report should be read with that limitation in mind.

Measurement Uncertainty

No numerical uncertainty budget is stated in this report: deriving one requires the deployment geometry and meteorological conditions, which are held with the study records rather than in the data file. For context, ISO 1996-2 notes that the combined standard uncertainty of an environmental noise measurement is typically of the order of 1 to 3 dB once instrument tolerance, microphone position, source variability and meteorological conditions are accounted for. Differences between values in this report smaller than that should not be treated as meaningful, and any comparison close to a guideline or limit should be interpreted accordingly rather than as a definitive pass or fail.

Source Attribution

Acoustic sensors measure total environmental energy; they do not definitively identify specific noise sources (e.g., distinguishing a commercial aircraft from a heavy goods vehicle). A source-specific comparison requires cross-referencing with external databases (e.g., ADS-B flight tracking).

Health vs. Legal Limits

This report evaluates data against both biological health guidelines (WHO 2018) and local regulatory limits (e.g., Maryland COMAR). A level at or below a regulatory limit does not guarantee the absence of adverse health effects.

Equipment Calibration

The accuracy of these metrics is strictly dependent on the proper deployment, windshielding, and recent acoustic calibration of the monitoring hardware.

Not Medical Advice

This data is for environmental and epidemiological research purposes. It does not constitute a clinical medical diagnosis or formal legal counsel.